

BA Topics Details

Monday, 7 September 2026 9:34 pm

Research or examine the following key items:

capabilities, business processes (define all three levels), **user personas, user stories**, and **requirements**.

These concepts are commonly used in **Business Analysis, BPM (Business Process Management), Agile, and System Development**. They are related but focus on different levels of detail.

1. Capabilities

A **business capability** describes **what an organization is able to do** to achieve its goals, regardless of how it is done, who does it, or what technology is used.

Characteristics:

- Stable over time
- Focuses on outcomes, not procedures
- Technology-independent
- High-level view of the business

Example: Uniqlo Click & Collect

Capability	Description
Order Management	Ability to receive and manage customer orders
Inventory Management	Ability to track and control stock levels
Payment Processing	Ability to accept and process payments
Customer Communication	Ability to notify customers about order status
Store Fulfilment	Ability to prepare orders for pickup

Notice that a capability does **not** describe the steps involved.

Capability = What the business can do

2. Business Processes

A **business process** describes **how work is performed** to achieve a business outcome.

Process = How the capability is delivered

Three Levels of Business Processes

Level 1: Process Area (Value Chain Level)

This is the highest level.

It shows major business functions.

Example: Uniqlo

Customer Shopping
├─ Online Ordering
├─ Inventory Management
├─ Fulfilment
├─ Payment Processing
└─ Customer Support

Purpose:

- Executive overview
- Strategic planning

Level 2: End-to-End Business Process

Shows the major activities of a process.

Example: Click & Collect

Customer places order

↓
Verify payment
↓
Reserve inventory
↓
Prepare order
↓
Notify customer
↓
Customer collects item

Purpose:

- Process understanding
- Cross-department communication

Level 3: Detailed Workflow / Task Level

Shows detailed tasks, decisions, systems, and roles.

Example: Reserve Inventory

System receives order
↓
Check stock availability
↓
Stock available?
├─ Yes → Reserve item
└─ No → Notify customer
↓
Update inventory database

Purpose:

- Staff training
- System design
- Automation

Relationship

Capability
↓
Level 1 Process
↓
Level 2 Process
↓
Level 3 Workflow
Example:
Capability:
Inventory Management
Level 1:
Inventory Operations
Level 2:
Reserve Stock for Click & Collect
Level 3:
Check stock → Reserve stock → Update database

3. User Personas

A **user persona** is a fictional representation of a group of users.

It helps teams understand:

it helps teams understand:

- Who uses the system
- Their goals
- Their needs
- Their frustrations

Example Persona

Sarah, 28

- Full-time office worker
- Shops online frequently
- Limited time during weekdays
- Wants fast pickup after work

Goals

- Buy clothes quickly
- Avoid delivery delays

Pain Points

- Long queues
- Out-of-stock items

Personas help teams design products around real user needs.

4. User Stories

A user story describes a requirement from a user’s perspective.

Standard Format

As a <user>,
I want <goal>,
So that <benefit>.

Examples

Customer

As a customer,
I want to collect my order from a nearby store,
So that I can avoid delivery fees.

Store Staff

As a store employee,
I want to view pending pickup orders,
So that I can prepare them quickly.

Administrator

As an administrator,
I want inventory updates to happen automatically,
So that stock levels remain accurate.

User stories focus on **user value**, not technical implementation.

5. Requirements

Requirements describe what the system or business solution must do.
They are more detailed than user stories.

Types of Requirements

Business Requirements

High-level business objectives.
Example:
The company shall provide a Click & Collect service
to increase customer convenience and reduce delivery costs.

Stakeholder Requirements

Needs of stakeholders.
Example:
Customers must be able to select a pickup store.
Store staff must receive pickup notifications.

Functional Requirements

What the system must do.
Example:
FR-001:
The system shall allow customers to choose a pickup store.
FR-002:
The system shall reserve inventory when payment is successful.
FR-003:
The system shall send an email notification when an order is ready.

Non-Functional Requirements

How well the system performs.
Example:
NFR-001:
The system shall process orders within 3 seconds.
NFR-002:
The system shall be available 99.9% of the time.
NFR-003:
Customer data shall be encrypted.

Complete Example (Hierarchy)

Capability

Order Management

Level 1 Process

Online Shopping

Level 2 Process

Click & Collect Order Fulfilment

Level 3 Process

- Receive Order
- Verify Payment
 - Reserve Stock
 - Prepare Order
 - Notify Customer
 - Pickup Confirmation

User Persona

Sarah, busy office worker

User Story

As a customer,
I want to collect my order from a nearby store,
So that I can receive my purchase faster.

Functional Requirement

The system shall allow customers to select a pickup location during checkout.

Non-Functional Requirement

The checkout page shall load within 2 seconds.

Simple Hierarchy

Business Goal

↓

Capability (What)
 ↓
 Business Process (How)
 ↓
 User Persona (Who)
 ↓
 User Story (Need)
 ↓
 Requirements (Detailed Specifications)
 ↓
 System Design & Development

This hierarchy is commonly used by Business Analysts to move from a **business objective** down to detailed system requirements that developers can build.

Understanding how these elements are interconnected is crucial. Write examples of each topic and explain to team

- Explain and examples of each topic
- Business Process Model and Notation (BPMN) Diagram
 - Use Case Diagram
 - Entity-Relationship (ER) Diagram
 - Data Flow Diagram (DFD)
 - Wireframe or Mockup Diagram
 - Swimlane Diagram (Cross-Functional Flowchart)
 - Gantt Chart
 - Stakeholder Map
 - State Diagram
 - Requirement Traceability Matrix (RTM)
 - Gap Analysis Diagram
 - Activity Diagram

1. Business Process Model and Notation (BPMN) Diagram

Purpose
 Shows how a business process works from start to finish.

Answers
 "How does the business operate?"

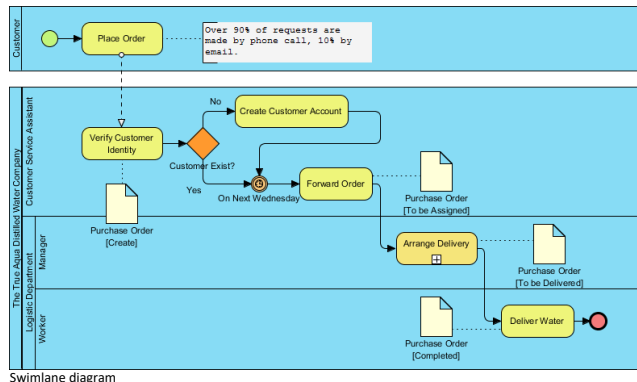
Example (Uniqlo Click & Collect)

(Start)
 ↓
 Customer places order
 ↓
 Payment successful?
 └─ No → Cancel Order
 └─ Yes
 ↓
 Reserve Inventory
 ↓
 Prepare Order
 ↓
 Notify Customer
 ↓
 Customer Collects Item
 ↓
 (End)

Used By

- Business Analysts
- Process Analysts
- Operations Teams

Output
 Understanding of current and future business processes.



Swimlane diagram

2. Use Case Diagram

Purpose
 Shows interactions between users and the system.

Answers
 "Who uses the system and what can they do?"

Example

Customer

- Place Order
- Track Order
- Collect Order

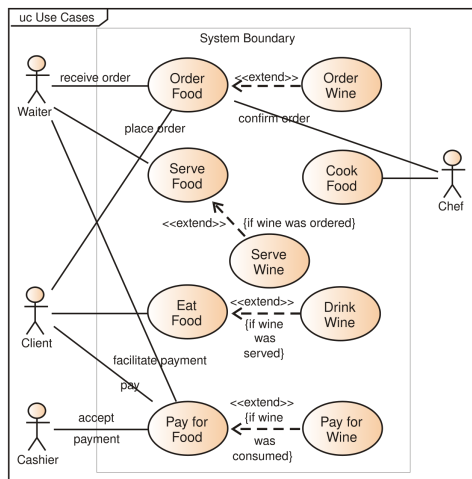
Store Staff

- Prepare Order
- Confirm Pickup

Used By

- Business Analysts
- Developers
- Stakeholders

Output
 System functionality from the user's perspective.



3. Entity-Relationship (ER) Diagram

Purpose
 Shows how data entities relate to each other.

Answers
 "What data do we need?"

Example

Customer

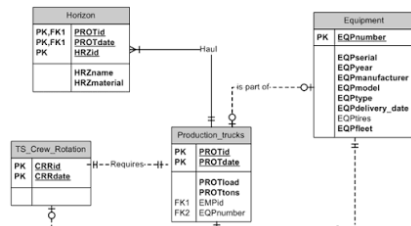
- 1
- M

Order

- M
- 1

Product

Example Entities



- Customer**
- CustomerID
 - Name
 - Email
- Order**
- OrderID
 - OrderDate
- Product**
- ProductID
 - Price
- Used By**
- Database Designers
 - Developers

Output
Database structure.



4. Data Flow Diagram (DFD)

Purpose
Shows how data moves through a system.

Answers
"Where does the data come from and where does it go?"

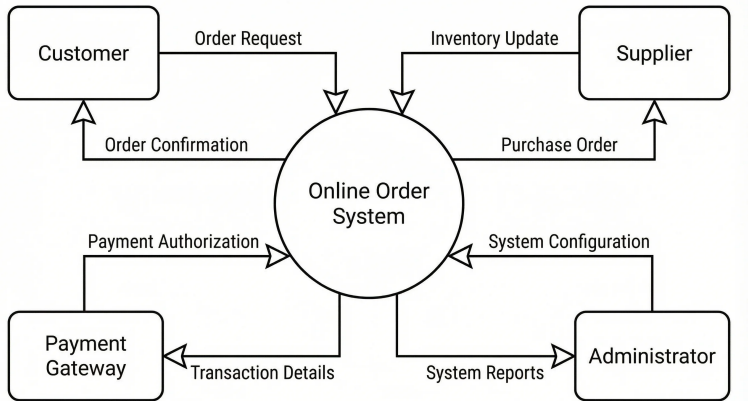
Example
Customer
↓
Place Order
↓
Order System
↓
Inventory Database

Order System
↓
Notification Service
↓
Customer

Used By

- Business Analysts
- Solution Architects

Output
Data movement and integrations.



5. Wireframe / Mockup Diagram

Purpose
Shows what the user interface will look like.

Answers
"What will users see?"

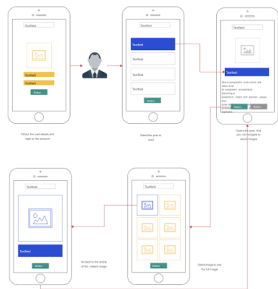
Example

UNIQLO
Product:
[T-Shirt]
Quantity: [1]
Pickup Store:
[Darwin CBD ▼]
[Place Order]

Used By

- UX/UI Designers
- Product Owners

Output
Screen designs.



6. Swimlane Diagram (Cross-Functional Flowchart)

Purpose
Shows process flow across different roles or departments.

Answers
"Who is responsible for each task?"

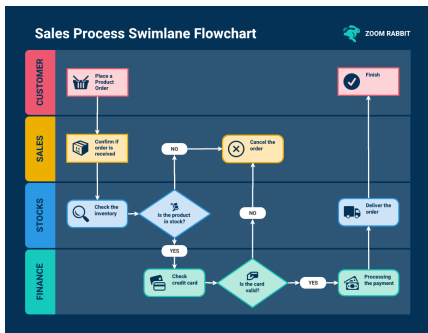
Example

Customer	System	Store Staff
Place Order		
	Process Payment	
	Reserve Stock	
		Prepare Order
	Send Notification	
Collect Item		Confirm Pickup

Used By

- Business Analysts
- Process Improvement Teams

Output
Role responsibilities.



7. Gantt Chart

Purpose
Shows project schedule and timeline.

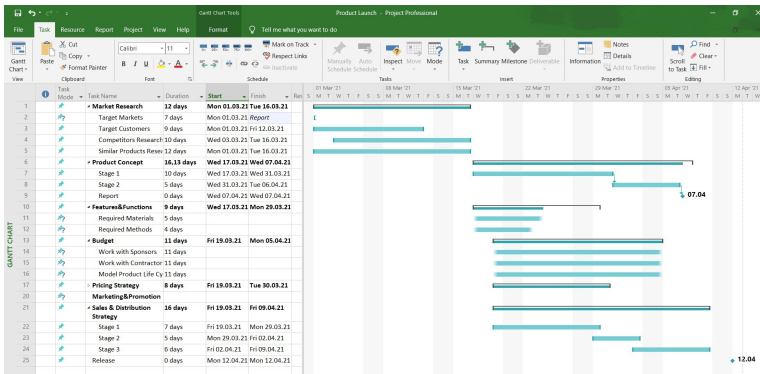
Answers
"When will work be completed?"

Example
Task
Requirements
Design
Development
Testing
Deployment

Used By

- Project Managers

Output
Project plan.



8. Stakeholder Map

Purpose
Identifies stakeholders and their influence.

Answers
"Who should we engage?"

Example
High Influence



Used By
• Business Analysts
• Project Managers

Output
Stakeholder engagement strategy.

9. State Diagram

Purpose
Shows how an object changes state.

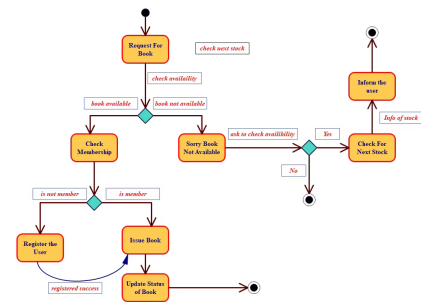
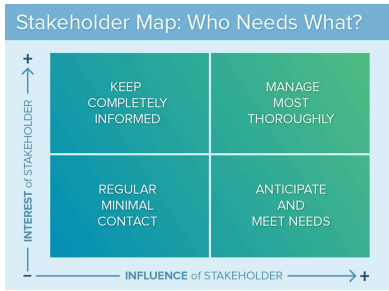
Answers
"What statuses can something have?"

Example (Order Status)

Created
↓
Paid
↓
Reserved
↓
Ready for Pickup
↓
Collected

Used By
• System Analysts
• Developers

Output
Lifecycle rules.



10. Requirement Traceability Matrix (RTM)

Purpose
Links requirements to design, development, and testing.

Answers
"Has every requirement been implemented and tested?"

Example

Req ID	Requirement	Design	Test Case
FR001	Select Store	UI-01	TC-01
FR002	Reserve Stock	API-01	TC-02
FR003	Notify Customer	API-02	TC-03

Used By
• Business Analysts
• Testers

Output
Requirement coverage.

Requirement Traceability Matrix							
Project Name: E-commerce Application							
Project ID: 112		Business Requirements Document (BRD)		Functional Requirements Document (FSD)		Test Case Document	
BR_ID	BR_User Case	FR_ID	FR_User Case	Priority	Test Case ID	Status	Comments
BR_1	Product Listing	FR_1	Sort by	High	TC_001	Finished	Dec 1: Testing started
					TC_002		Dec 6: Defect reported
BR_2	Payment Module	FR_2	Filters	High	TC_003	Finished	Dec 12: Defect Fixed
					TC_004		Dec 15: FS_Passed
BR_3	Payment Module	FR_3	By Credit Card	High	TC_005	In Progress	Dec 1: Testing started
					TC_006		Dec 1: Testing started
BR_4	Payment Module	FR_4	By Debit Card	High	TC_007	Not Started	Dec 1: Testing started
					TC_008		Dec 1: Testing started

11. Gap Analysis Diagram

Purpose
Compares current state and future state.

Answers
"What needs to change?"

Example

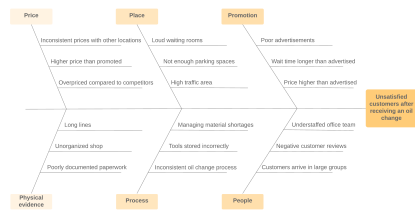
Current State	Future State	Gap
Manual stock update	Real-time inventory	Automation required
Email orders	Mobile app orders	New application needed
Phone pickup confirmation	Automated notification	Notification system needed

Used By
• Business Analysts
• Management

Output
Improvement roadmap.
<https://www.projectmanager.com/blog/gap-analysis-project-management>



SWOT Diagram



Fishbone Diagram
The most commonly used categories for investigation are:

- Measurements
- Materials
- People
- Machines
- Methods
- Environment

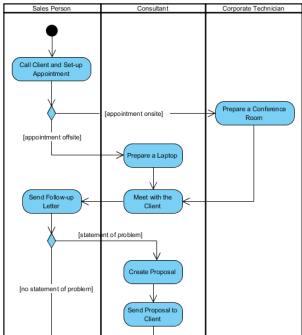
12. Activity Diagram

Purpose
Shows workflow logic and decisions in a process.

Answers
"What actions occur and what decisions are made?"

Example

Start
↓
Place Order
↓
Payment Successful?
└─ No → Cancel Order
└─ Yes
↓
Reserve Inventory
↓
Notify Customer
↓
End



Used By

- Business Analysts
- System Analysts
- Developers



Output

Detailed process logic.

Summary Table

Diagram	Main Question
BPMN	How does the business process work?
Use Case	Who uses the system and what do they do?
ER Diagram	What data is needed?
DFD	How does data flow?
Wireframe	What will the screen look like?
Swimlane	Who performs each activity?
Gantt Chart	When will work be done?
Stakeholder Map	Who should be engaged?
State Diagram	What states can an object have?
RTM	Are all requirements implemented and tested?
Gap Analysis	What changes are needed?
Activity Diagram	What workflow and decisions occur?

End-to-End Example (Uniqlo Click & Collect)

1. **Stakeholder Map** → Identify customer, store staff, manager, IT team.
2. **BPMN/Activity Diagram** → Model the Click & Collect process.
3. **Swimlane Diagram** → Show responsibilities across customer, system, and store staff.
4. **Use Case Diagram** → Define system interactions.
5. **Wireframes** → Design order and pickup screens.
6. **DFD** → Show order and inventory data flow.
7. **ER Diagram** → Design customer, order, and product tables.
8. **State Diagram** → Define order status transitions.
9. **Gap Analysis** → Compare current vs future process.
10. **RTM** → Ensure every requirement is built and tested.
11. **Gantt Chart** → Plan and track project delivery.

This sequence demonstrates how all these modelling techniques work together from **business understanding** → **requirements** → **design** → **implementation** → **testing**.